

Same Goal, Different Words: Spatial Instruction Sensitivity in World–Action Models

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Abstract: A robot should respond to the requested physical goal, not to one favored way of describing it. We test this requirement by asking three world–action models to perform the same placement while describing the spatial relation from either the moved object’s or the reference object’s perspective. Across 864 matched rollouts on four RoboLab scenes, reference-first descriptions produce fewer stable spatial outcomes, with an average gap of 12.2 percentage points. This gap is concentrated in particular relations rather than shared uniformly across scenes. We also find that briefly satisfying a spatial relation can conceal later failure or movement of the wrong object. These findings motivate a more discriminating evaluation of world–action models: test alternative descriptions of the same goal, examine individual relations, and distinguish reaching a spatial arrangement from completing the instruction.

Keywords: World–Action Models, Spatial Language, Robot Evaluation

1 Introduction

Consider asking a robot to place a mustard bottle to the right of a raisin box. The instruction can describe the bottle as being to the right of the box, or the box as being to the left of the bottle. The required arrangement is unchanged, and both instructions can explicitly identify the bottle as the object to move. If the resulting behavior changes, physical task difficulty alone cannot explain the difference: the robot started in the same scene and was asked to achieve the same goal.

This is a useful test for world–action models, which generate actions alongside predictions of future observations. For robot control, predictive capability should be assessed alongside reliable goal conditioning. We ask: *when the physical goal and initial state stay fixed, does describing the relation from the other object’s perspective change what the model accomplishes?*

Instruction sensitivity is already established in robot learning. LIBERO-Plus evaluates language perturbations [1], LIBERO-Para tests meaning-preserving paraphrases from shared starts [2], and semantic guidance for world–action models targets instruction alignment [3]. Our contribution is a focused comparison of relational reference inversion across three world–action policies in existing RoboLab scenes [4]. As in controlled simulation evaluations [5], shared starts help separate the intervention from scene difficulty.



Same requested placement; same object to move

Mover-first (S): Move the mustard bottle on the table so that the mustard bottle is to the right of the raisin box.

Reference-first (I): Move the mustard bottle on the table so that the raisin box is to the left of the mustard bottle.

Figure 1: Exact tested mustard-right prompts and two exterior views of the asset from a pilot reset. Confirmation pairs share a frozen start; policies also receive a wrist view. The intervention changes argument order and relation words, not merely token order.

2 Paired spatial evaluation

Change the description, hold the goal fixed. We use four RoboLab scenes: a cube and bowl, butter and a raisin box, mustard and a raisin box, and two bowls. Goals cover left/right, front/behind, supported placement on top, and stacking either bowl on the other. Seven goals are native to their scenes; five are added conditions on existing assets. A planned two-bin scene failed scripted feasibility qualification and was excluded before confirmation, independently of learned-policy outcomes.

Every goal has three descriptions: direct wording (D), native where available; an expanded description that names the moved object first in the relation (S); and a description that names the reference object first and reverses the relation (I). Figure 1 shows an exact pair. Both forms explicitly request movement of the same object. For supported placement, “on top of and supported by” becomes “underneath and supporting.” Bowl wording is less exact: “underneath and supporting” does not explicitly require the nesting imposed by its scorer. Its results are shown separately.

Compare models from the same starts. We evaluate the DROID checkpoints of Cosmos3 Nano [6], Cosmos3 Edge [7], and FLUX 3 Action [8]. All receive the native wrist and two exterior cameras plus proprioception. We use eight frozen starts per scene, selected before confirmation as the first controller-qualified proposals under small position and yaw perturbations. Each model runs every goal and description from those starts, yielding 864 confirmation episodes; development runs are excluded. Initial-state and first-observation hashes match across paired descriptions.

Policies execute 32-action chunks at 15 Hz, with fixed inference seed 6100 and checkpoint-specific inference settings. Episodes run for their full scene-specific horizon (20–40 s), without stopping at the first successful placement. Exact prompts, settings, states, and receipts accompany the paper.

Separate reaching a placement from completing the task. The prespecified behavioral measure records whether the requested relation, support, and gripper detachment hold for one continuous second while the designated mover remains below 0.02 m/s. Relations are evaluated in the robot frame. We call this a *stable spatial outcome*. It can occur at any point in the episode. We separately check whether it remains stable during the final second; this terminal check is post-hoc. Crucially, the relation score does not verify which object the robot moved. Moving the reference may also make the relation true, so a score pass is not sufficient evidence that the instruction was followed.

Goals already satisfied at reset are separated from new-goal achievement using the frozen state registry. This corrects inconsistent initial labels in 44 episode summaries without changing their recorded outcomes. Three goals are initially satisfied at every start: cube-right, butter-right, and mustard-left. Our prespecified wording contrasts average paired differences equally across models, scenes, and goals within scene. Confidence intervals resample physical starts within each scene, keeping their associated models, goals, and descriptions together; paired cluster sign-flip tests use Holm correction for the two contrasts.

3 Results

Reference-first descriptions produce fewer stable outcomes. The mover-first form reaches the stable spatial endpoint more often than the reference-first form, with a weighted gap of 12.2 percentage points (95% interval: 8.6–15.4; adjusted $p < 0.001$). The average has the same sign for all three models (Figure 2), while the direct-versus-expanded contrast is smaller and inconclusive. Not every rephrasing demonstrably reduces performance. The post-hoc terminal check retains a similar reference-inversion gap, so the finding is not explained solely by transient placements.

A few relations drive the average. The largest gap occurs when placing mustard to the right of the raisin box. Reference inversion sharply reduces stable outcomes for this goal across the three models. In contrast, the cube scene has no aggregate S/I difference, and cube-behind actually favors the reference-first form under the recorded metric. Figure 3 exposes these differences: performance

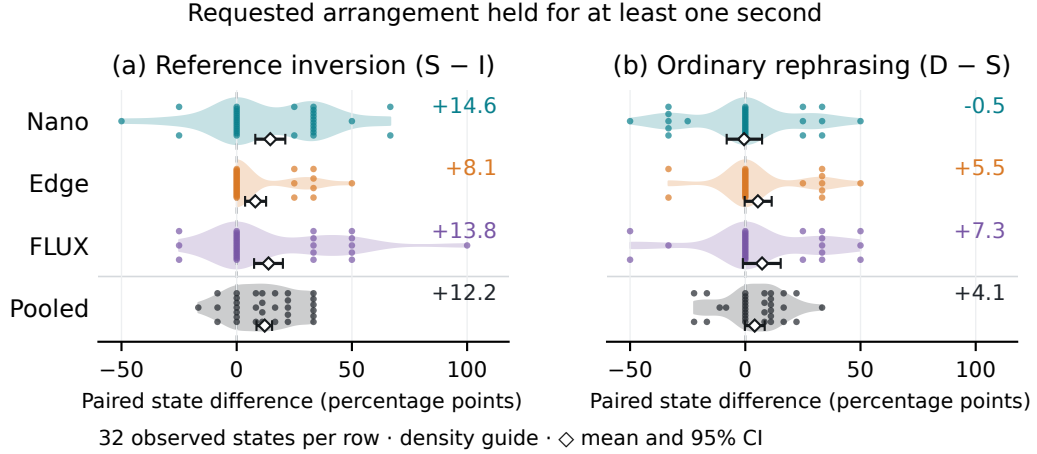


Figure 2: Paired wording effects. Positive S–I favors mover-first descriptions; D–S compares direct and expanded wording. Each dot averages paired differences over goals within one start (also over models in the pooled row). Shading guides the eye; diamonds and bars show means and 95% state-bootstrap intervals. Model-level intervals are descriptive.

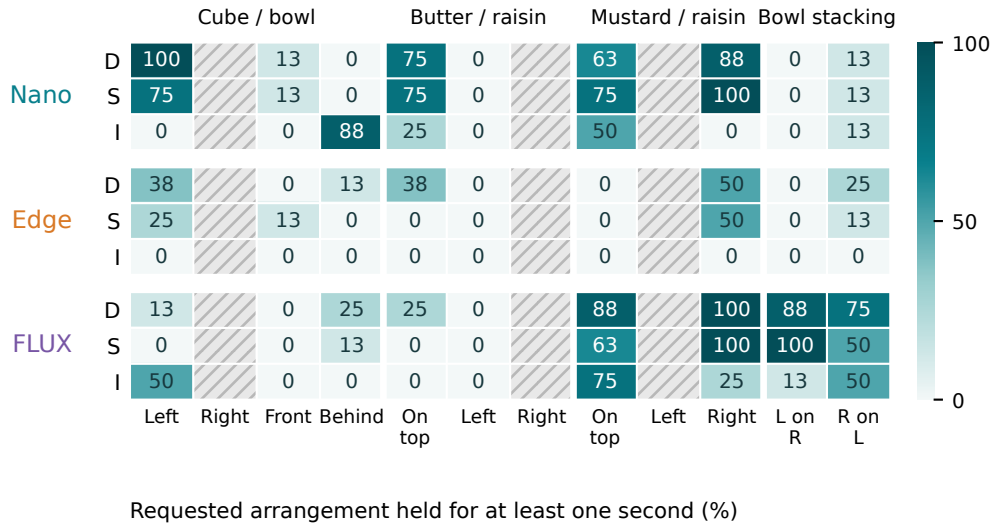


Figure 3: Holding the requested arrangement, by goal and wording. Each colored cell has eight starts; hatching marks initially satisfied goals. Relation passes do not verify complete instruction execution, and bowl wording has a support/containment ambiguity.

varies with the particular scene, relation, and model. The evidence supports a weakness in these tested combinations, rather than a universal inability to interpret left, right, or reversed references.

The success definition changes the interpretation. Roughly half of the episodes that reach a stable outcome lose stability before termination. Counting the first stable placement therefore gives a much more favorable picture than checking how the episode finishes. The same distinction applies to object roles. All seven reference-first cube-behind passes come from Nano episodes with no detected cube lift and substantial bowl displacement. The score has registered the requested relation, but these summaries do not establish that the robot moved the cube as instructed. Relation scores alone cannot distinguish faithful execution from moving the reference into a favorable arrangement.

4 Discussion and scope

For the question “Do robots need world models?”, our findings motivate a concrete evaluation requirement. A model’s actions should be tested under alternative descriptions of the same physical goal, with results broken down by relation and object role. Benchmark reports should distinguish reaching a relation, retaining it, and manipulating the intended object. These checks complement evaluations of predictive quality; our experiment does not compare otherwise matched systems with and without a world model.

The findings concern three checkpoints, four assets, a small set of controller-qualified starts, one inference seed, and the tested phrases. Reference inversion changes vocabulary and argument order together, and the bowl descriptions are not fully equivalent to their containment scorer. Broader claims require more linguistic variation and scoring that explicitly verifies the requested manipulation.

The original protocol’s primary question concerned forecast-based failure diagnosis; the wording comparisons were prespecified secondary analyses. After behavioral outcomes were available, automated vision-language annotators replaced planned human annotations. Their poor agreement and lack of detectable diagnostic improvement leave forecast informativeness unresolved. We center this paper on the behavioral evidence and retain that original analysis and amendment with the artifacts. The measured wording sensitivity does not establish that world predictions are unnecessary or uninformative.

References

- [1] S. Fei, S. Wang, J. Shi, Z. Dai, J. Cai, P. Qian, L. Ji, X. He, S. Zhang, Z. Fei, J. Fu, J. Gong, and X. Qiu. LIBERO-Plus: A progressive robustness benchmark for visual-language-action models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 38574–38583, 2026.
- [2] C. Kim, M. Kim, M. Kang, H. Kim, and D. Jung. LIBERO-Para: A diagnostic benchmark and metrics for paraphrase robustness in VLA models. *arXiv preprint arXiv:2603.28301*, 2026. URL <https://arxiv.org/abs/2603.28301>. Accepted to EMNLP 2026, main conference.
- [3] J. He, J. Li, Z. Zhong, H. Yan, R. Li, Y. Zheng, J. Zhu, T. Zhang, Y. Du, W. Chen, S. Zhou, and H. Li. SG-WAM: Text-grounded and spatial-aware semantic guidance for world-action models. *arXiv preprint arXiv:2608.08839*, 2026. URL <https://arxiv.org/abs/2608.08839>.
- [4] X. Yang, R. Dagli, A. Zook, H. Hadfield, A. Goyal, S. Birchfield, F. Ramos, and J. Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, 2026. doi:10.15607/RSS.2026.XXII.096.
- [5] X. Li, K. Hsu, J. Gu, O. Mees, K. Pertsch, H. R. Walke, C. Fu, I. Lunawat, I. Sieh, S. Kirmani, S. Levine, J. Wu, C. Finn, H. Su, Q. Vuong, and T. Xiao. Evaluating real-world robot manipulation policies in simulation. In *Proceedings of the 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 3705–3728. PMLR, 2025.
- [6] NVIDIA. Cosmos3-Nano-Policy-DROID. Hugging Face model card and checkpoint, 2026. URL <https://huggingface.co/nvidia/Cosmos3-Nano-Policy-DROID>. Accessed September 27, 2026.
- [7] NVIDIA. Cosmos3-Edge-Policy-DROID. Hugging Face model card and checkpoint, 2026. URL <https://huggingface.co/nvidia/Cosmos3-Edge-Policy-DROID>. Accessed September 27, 2026.
- [8] Black Forest Labs. FLUX 3 Action DROID. Hugging Face model card and checkpoint, 2026. URL <https://huggingface.co/black-forest-labs/flux-3-action-droid>. Accessed September 27, 2026.